ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

Kenan Song
School of Chips
Email:
Kenan.Song@xjtlu.edu.cn
Office: C5007

Density of State I

I. Brief Introduction to Density of State (DOS)

II. Van Hove Singularity-Origin of Topology

III. Fermi Dirac Distribution and DOS

IV. DOS related Properties

I. Brief Introduction to Density of State (DOS)

The Microscopic Problem: An Avogadro's number (10^{23}) of electrons occupying discrete quantum states described by wavevectors (k)

The Macroscopic Target: Calculating total carrier density (*n*), internal energy (*U*), or electronic specific heat (C_v)

The Bottleneck: Tracking every single state $\Psi_{n,k}$ is computationally and conceptually impossible

The Solution: Condense all *k*-space information into a single energy axis variable: *g*(*E*)

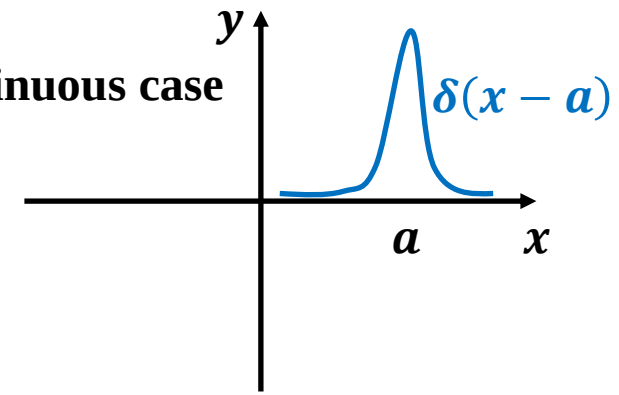
Think about the sheer complexity of a real crystal. There are trillions of electrons moving in a periodic potential. If one wants to compute a thermodynamic property, one should not care where in the Brillouin zone an electron sits or what its exact momentum vector is; one only care about its energy. DOS acts as a statistical counting mechanism. It compresses a multi-dimensional reciprocal space into a simple, one-dimensional function of energy

I. Brief Introduction to Density of State (DOS)

DOS — The number of available quantum states per unit energy interval per unit volume of the crystal

$$g(E) = \frac{1}{V}\sum_{n,k} \delta[E - E_n(k)]$$

Continuous case



Discrete case

$$\delta(x-a) = \begin{cases} \infty & x = a \\ 0 & x \neq a \end{cases}$$

where, *V* is the crystal volume, *n* is the band index, *k* is the wavevector, and δ is the Dirac Delta function

Units: States/(eV·cm^3) in 3D systems

The Dirac delta function $\delta[E - E_n(k)]$ acts as a rigorous 'quantum filter' or 'sieve'. It scans across all bands *n* and all allowed wavevectors *k* in the first Brillouin Zone. If the eigenvalue $E_n(k)$ matches the probe energy *E*, the delta function registers a hit. Sum these hits up and normalize by the volume. Notice that this formula assumes discrete states, which is exactly how to handle finite *k* point meshes in numerical codes before applying broadening

I. Brief Introduction to Density of State (DOS)

In an infinite crystal, *k* points become quasi-continuous $\sum_{k} \cdots \rightarrow \frac{V}{(2\pi)^d} \int d^d k$

***d* - dimensionality**

***γ* represents internal degrees of freedom (*e.g.*, spin degeneracy *γ* = 2)**

$$g(E) = \frac{\gamma}{(2\pi)^3} \int_{BZ} \delta[E - E_n(k)] d^3 k$$

When transiting from a finite molecule to a bulk periodic crystal, the discrete spacing between *k* points approaches zero. This allows one to convert the sum into a continuous volume integral over the Brillouin Zone. The factor $(2\pi)^3/V$ is the primitive volume of a single *k* point in reciprocal space. Do not forget the spin degeneracy factor *γ*.

For non-magnetic systems, it is simply 2, but when working on spintronics or spin-polarized materials, this degeneracy splits, requiring spin-up and spin-down DOS to be treated independently

I. Brief Introduction to Density of State (DOS)

Transforming the volume element: $d^3k = dS_k dk_\perp$

Since $dE = |\nabla_k E| dk_\perp$**, one can substitute** $dk_\perp$

The Geometrical DOS Formula: $$g(E) = \frac{\gamma}{(2\pi)^3} \int_{s(E)} \frac{dS_k}{|\nabla_k E|}$$

where, $S(E)$ is the constant-energy surface in k space, and $\nabla_k E$ is the energy gradient

This is one of the most elegant transformations in solid-state physics. Rewrite the 3D volume integral by slicing it into nested shells of constant energy surfaces, $S(E)$. The thickness of the shell is $dk_\perp$. By using the chain rule, one relate this thickness directly to the gradient of the band energy, $|\nabla_k E|$. Look at where the gradient is located—it is in the denominator. This tells that the shape of the DOS is fundamentally dictated by how fast or slow the bands change with respect to k

I. Brief Introduction to Density of State (DOS)

Semiclassical Electron Velocity: $V_g(k) = \frac{1}{\hbar}\nabla_k E(k)$

Inverse Proportionality: $g(E) \propto \oint \frac{dS_k}{|V_g|}$

Physical Consequences:

Dispersive (Curved) Bands: High velocity → Electrons move fast → Low DOS

Flat Bands: Zero or low velocity → Electrons are localized/heavy → Massive spikes in DOS

Connect this to real-world electronics. $\frac{1}{\hbar}\nabla_k E$ is nothing but the semiclassical group velocity V_g of the electron. Therefore, the density of states at a given energy is inversely proportional to how fast electrons can travel at that energy. If a band is highly dispersive—meaning it climbs sharply like a steep hill—electrons pass through that energy range quickly, leading to a low DOS. Conversely, if one has localized *d* or *f* orbitals, or flat bands generated by Moire superlattices or heavy-fermion systems, the gradient drops to zero. The electrons stall, stacking up at that exact energy level, creating giant peaks in the DOS plot

II. Van Hove Singularity-Origin of Topology

Non-analytic mathematical features (kinks, peaks, steps) in the DOS arising from critical points where $\nabla_k E = 0$

Types of Critical Points in 3D:

M_0: Local Minimum (band edge)

M_1, M_2: Saddle Points (minimax points)

M_3: Local Maximum

Theorem: Leon Van Hove (1953) proved that periodic lattices must possess these singularities due to topology

When the energy gradient $\nabla_k E$ vanishes exactly, the geometric integral encounters a denominator of zero. These specific locations in the Brillouin Zone are called critical points, and the features they carve into the DOS are known as Van Hove Singularities.

Leon Van Hove proved topologically that because the Brillouin Zone is a closed, periodic torus, these critical points are mathematically guaranteed to exist. They are not anomalies or bugs in the calculation; they are fundamental signatures of the lattice symmetry itself

II. Van Hove Singularity-Origin of Topology

3D Bulk: Continuous, unconfined motion in all directions (*x*, *y*, *z*)

2D Quantum Well/Monolayer: Confinement along *z*; free motion in *x*, *y* (e.g., Graphene, MoS_2)

1D Quantum Wire/Nanotube: Confinement along *y*, *z*; free motion along *x*

0D Quantum Dot/Atom: Confinement in all three directions (*x*, *y*, *z*)

This a beautiful demonstration of quantum mechanics in action: dimensionality

By changing the physical constraints of a material—moving from bulk crystals to monolayers, nanowires, or isolated clusters—one fundamentally alter the available phase space in *k* space

This structural tailoring acts like a set of scissors, chopping up the smooth dispersion relations and forcing the DOS to adopt completely different mathematical profiles

II. Van Hove Singularity-Origin of Topology

3D Systems: Free Electron Approximation

Dispersion: $E(k) = \frac{\hbar^2(k_x^2+k_y^2+k_z^2)}{2m^*}$

***k* space volume enclosed by an energy sphere:** $\Omega_k = \frac{4}{3}\pi k^3 = \frac{4}{3}\pi\left(\frac{2m^*E}{\hbar^2}\right)^3$

3D DOS Derivation Result: $g_{3D}(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E}$

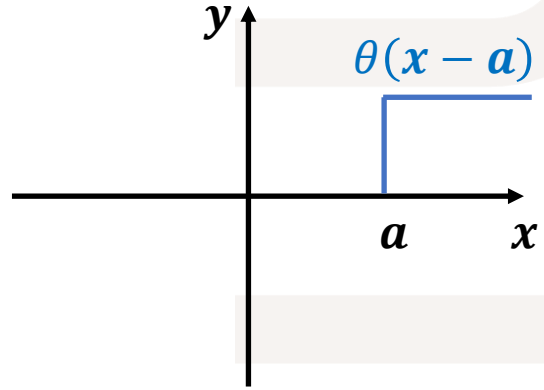
Feature: Smooth, continuous $\sqrt{E}$ onset from the band edge

Let's derive the classic 3D DOS using the free-electron model with an effective mass m^*. If the band is isotropic and parabolic, the constant-energy surface is a perfect sphere in *k* space. The volume of this sphere scales as k^3, which means it scales as $E^{3/2}$. To find the density of states, we take the derivative of this volume with respect to energy ($^{d\Omega_k}/_{dE}$). The derivative of $E^{3/2}$ gives us a function proportional to $E^{1/2}$ — or $\sqrt{E}$. This is why the DOS of any standard bulk semiconductor or metal rises smoothly like a square-root curve starting right at the band edge

II. Van Hove Singularity-Origin of Topology

2D Systems: Quantum Wells and Monolayers

Dispersion: $E(k) = \frac{\hbar^2(k_x^2+k_y^2)}{2m^*}$; **where,** E_n **represents quantized subband levels**



k **space area of a constant-energy circle:** $A_k = \pi k^2 = \pi \frac{2m^*(E-E_n)}{\hbar^2}$

2D DOS Derivation Result: $g_{2D}(E) = \frac{m^*}{\pi\hbar^2}\sum_n \theta(E - E_n)$; **where,** θ **is the Heaviside step function**

Feature: Energy-independent plateaus; sharp staircases

Now let us trap electrons inside a 2D plane, such as a quantum well or a van der Waals monolayer. Motion along the *z* axis is quantized into discrete energy levels E_n, while motion in the *x*-*y* plane remains free and parabolic. Instead of a 3D sphere, the constant-energy surface collapses into a 2D circle. The area of a circle scales as k^2, which is directly linear with energy *E*. When we take the derivative of a linear function with respect to *E*, we get a constant. This means the 2D DOS does not depend on energy at all within a subband. It turns into a series of sharp, flat steps governed by the Heaviside step function θ

II. Van Hove Singularity-Origin of Topology

1D Systems: Carbon Nanotubes and Nanowires

Dispersion: Only k_x remains continuous; y and z are strictly quantized

k space length of a 1D line segment: $L_k = 2k = 2\sqrt{\frac{2m^*(E-E_{n,m})}{\hbar^2}}$

1D DOS Derivation Result: $g_{1D}(E) = \frac{1}{\pi}\left(\frac{2m^*}{\hbar^2}\right)^{1/2} \sum_{n,m} \frac{1}{\sqrt{E-E_{n,m}}}$

Feature: Sharp, divergent peaks that drop off as energy increases $\frac{1}{\sqrt{E}}$

What happens if one confine electrons even further into a 1D wire? Now, only one direction of momentum is continuous. The available states are restricted to 1D line segments. The length of these segments scales as $k \propto \sqrt{E}$. When we take the derivative of $\sqrt{E}$ to get our DOS, the energy term moves to the denominator, yielding a $1/\sqrt{E}$ dependency. This creates a striking visual signature: right at the onset of each quantized subband $E_{n,m}$, the DOS shoots up to infinity, forming a highly divergent spike before decaying smoothly. These sharp features are incredibly useful for enhancing optical transitions in 1D lasers

II. Van Hove Singularity-Origin of Topology

0D Systems: Quantum Dots and Isolated Atoms

Dispersion: Completely broken. No continuous *k* vectors remain

All spatial dimensions are smaller than the electron's de Broglie wavelength

0D DOS Formulation: $g_{0D}(E) = \sum_i \gamma_i \delta(E - E_i)$

Feature: A series of discrete, ultra-sharp delta function lines. Often called 'Artificial Atoms'

Regarding the zero-dimensional limit—the quantum dot, the electron's ability to travel as a free wave is stripped. Because there is no continuous reciprocal space left to integrate over, the integral reverts to its purest, most primitive form: a sum over discrete delta functions.

The DOS looks identical to the sharp, discrete energy levels of an isolated single atom. This is why quantum dots are frequently called 'artificial atoms.' They exhibit delta-like absorption and emission profiles, which explain their exceptionally vibrant and pure colors in display technologies

II. Van Hove Singularity-Origin of Topology

Visualizing the Density of States Landscape

Dimension	Geometry	Mathematical Dependency	DOS Graphic Profile
3D	Bulk	$g(E) \propto \sqrt{E}$	Parabolic Curve
2D	Quantum Well	$g(E) \propto$ constant	Staircase Steps
1D	Nanowire	$g(E) \propto {}^{1}/_{\sqrt{E}}$	Divergent Spikes
0D	Quantum Dot	$g(E) \propto \delta(E)$	Discrete Lines

III. Fermi Dirac Distribution and DOS

Populating the States: The Fermi-Dirac Filter

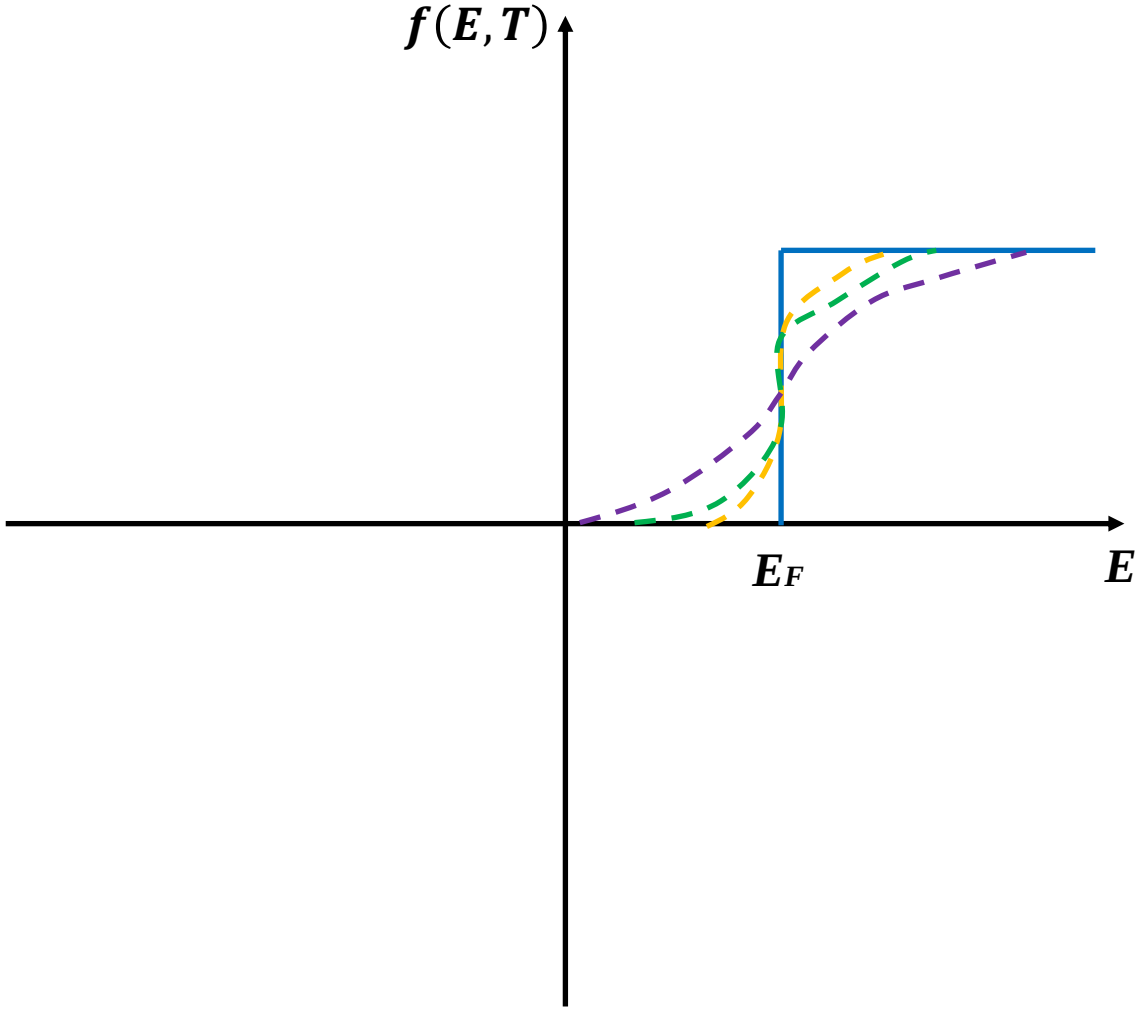
DOS, $g(E)$, only tells us where states exist; it does not tell us if they are occupied

The Occupation Function: Fermi-Dirac Distribution: $$f(E,T) = \frac{1}{e^{(E-\mu)/k_BT} + 1}$$

The Master Equation for Carrier Concentration: $$n = \int_{-\infty}^{\infty} g(E) f(E,T) dE$$

To connect the density of states to actual, measurable material properties, one has to introduce statistics. $g(E)$ gives us the structural blueprint of available 'parking spaces' for electrons. To know how many cars are actually parked, one must multiply $g(E)$ by the Fermi-Dirac distribution function, $f(E, T)$. At absolute zero, this distribution is a perfect step function, cutting off sharply at the Fermi level. As temperature rises, thermal energy smears out the edge, allowing electrons to spill over into higher available states. Integrating this product yields the total carrier density, n

III. Fermi Dirac Distribution and DOS



$$f(E,T) = \frac{1}{e^{(E-\mu)/k_BT} + 1}$$

$T_4 > T_3 > T_2 > T_1 = 0\ Kelvin$

IV. DOS related Properties

Electronic Diagnostics I: Metals and Phase Space Heat

Definition of a Metal: The Fermi level E_F intersects a continuous, non-zero region of the DOS, $g(E_F) > 0$

Electronic Specific Heat: $C_{el} = \frac{\pi^2}{3} k_B^2 T \cdot g(E_F)$

Pauli Paramagnetic Susceptibility: $\chi_P = \mu_B^2 \cdot g(E_F)$

Key Insight: Only electrons within ~ $k_B T$ of the Fermi level can actively participate in thermal and magnetic excitations

Let us look at our first major material diagnostic: Metals

In a metallic system, the Fermi level settles right in the middle of a band, meaning $g(E_F)$ is non-zero. This has immense consequences. When heating a metal, the vast majority of electrons deep inside the Fermi sea cannot absorb energy because the states directly above them are already full. Only the tiny fraction of electrons sitting right at the surface—within a window of $k_B T$ around E_F—can jump into empty states. Consequently, macroscopic properties like electronic specific heat (C_{el}) and Pauli paramagnetism (χ_P) are directly proportional to the value of the DOS exactly at the Fermi level

IV. DOS related Properties

Electronic Diagnostics II: Semiconductors *v.s.* Insulators

The Band Gap Signature: A region where $g(E) = 0$ over a finite energy range E_g

Semiconductors: Small gap ($E_g < 3$ eV). Thermally activatable

Insulators: Large gap ($E_g > 3$ eV). Thermally inert

Valence Band: Filled states below the gap ($E < E_v$)

Conduction Band: Empty states above the gap ($E > E_c$)

When analyzing non-metallic systems, the defining feature of DOS plot is a completely flat line of zero states.

This represents the band gap (E_g). The states below this quiet zone form the Valence Band, which is completely filled with electrons at $T = 0$ K, while the states above form the Conduction Band, which is entirely empty. If the gap is narrow, like in Silicon or Gallium Arsenide, thermal energy at room temperature is enough to kick a few electrons across the gap. If the gap is exceptionally wide, the material remains a strict insulator because the cost to excite an electron is simply too high

IV. DOS related Properties

Projecting the DOS: Partial and Orbital Decomposition (PDOS)

Total DOS (TDOS): The sum of all electronic states across the entire system

Partial/Projected DOS (PDOS): Decomposing the total DOS onto specific atomic sites or localized orbital symmetries (*s*, *p*, *d*, *f*)

Mathematical Projection: $g_{\alpha,l}(E) = \sum_{n,k} \left| \langle \Phi_{\alpha,l} | \Psi_{n,k} \rangle \right|^2 \delta[E - E_n(k)]$

Utility: Identifies the chemical origin of specific bands and tracks orbital hybridization

Up until now, the Total DOS has been discussed. But if one wants to understand the chemistry of a material, Total DOS is not enough; one needs the Projected or Partial DOS (PDOS).

By projecting the global Bloch wavefunctions $\Psi_{n,k}$ onto a localized atomic orbital basis $\Phi_{\alpha,l}$ (like the p_z orbital of carbon or the *d* orbitals of a transition metal), one can see exactly which atoms and orbitals are responsible for which peaks. For instance, in spintronics or catalysis research, PDOS can pinpoint exactly which *d* orbital states are crossing the Fermi level, providing complete control over chemical tuning

IV. DOS related Properties

Real-World Numerical Issues — The Necessity of Broadening

The Numerical Challenge: Explicit *k* point sampling yields a collection of sharp spike lines, not a smooth curve

The Solution: Replace the ideal Dirac delta function with a smooth broadening function

Common Smeared Functions:

$$\text{Gaussian Smearing: } \delta(x) \approx \frac{1}{\sigma\sqrt{\pi}} e^{-(x/\sigma)^2}$$

$$\text{Lorentzian Smearing: } \delta(x) \approx \frac{1}{\pi} \frac{\Gamma}{x^2+\Gamma^2}$$

The Danger: Over-smearing erases physical Van Hove features; under-smearing causes artificial noise

When implementing code to compute the DOS—such as using a triangle or tetrahedron integration scheme—one runs into a practical challenge. Because the computer can only sample a finite grid of *k* points, the raw output will look like a jagged mess of discrete spikes rather than a smooth, continuous physical curve. To fix this, implement numerical broadening. Swap out the mathematical Dirac delta function for a Gaussian or Lorentzian distribution with a width parameter (σ or Γ). However, one must find the right balance: if the broadening parameter is too small, DOS will be full of artificial noise; if it is too large, one may wash away the real, physical Van Hove singularities



That's all for this session

Thank you for your attention and great participation